

11-Naive_Bayes

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1 Naive Bayes Classifier

The **Naive Bayes classifier** is a family of probabilistic classifiers that rely on Bayes' Theorem and the assumption of feature independence. The classifier is called naive because it assumes that all the features in a dataset are independent of each other, which is rarely true in real-world scenarios, but it simplifies the computation.

Naive Bayes works by calculating the probabilities for each class and then predicting the class with the highest probability based on the input features. It is a powerful and simple algorithm for classification tasks, especially in domains like text classification (e.g., spam detection, sentiment analysis).

When to Use Naive Bayes Classifier? Naive Bayes is particularly useful in the following scenarios:

- **When working with high-dimensional datasets:** It performs well with many features, such as in text classification where each word can be a feature.
- **When the assumption of conditional independence holds:** Even when this assumption is not completely true, Naive Bayes often performs surprisingly well in practice.
- **For real-time predictions:** Naive Bayes is computationally efficient and can handle large datasets quickly, making it suitable for real-time applications.
- **Text classification, sentiment analysis, and spam filtering:** These tasks involve high-dimensional and sparse data, where Naive Bayes shines.
- **Categorical input features:** The algorithm handles categorical variables well, especially in its Multinomial Naive Bayes variant.

Who Should Use Naive Bayes Classifier? Naive Bayes is an excellent choice for:

- **Data scientists and machine learning practitioners** who need a fast, simple, and interpretable model for classification tasks.
- **Business analysts** looking for text-based solutions, such as customer sentiment analysis, email spam filtering, or document categorization.
- **Researchers and developers** working with large-scale text processing, like in Natural Language Processing (NLP).
- **Beginners in machine learning**, as Naive Bayes is easy to implement and understand, while still being effective in many real-world scenarios.

Advantages of Naive Bayes Classifier:

- **Fast and efficient:** Naive Bayes is computationally efficient and works well with large datasets.
- **Handles high-dimensional data:** Especially useful for text classification problems where the data is sparse and high-dimensional.
- **Simple to implement:** The algorithm is simple to code and interpret, making it a good starting point for beginners.
- **Requires less training data:** It performs well with relatively small datasets because it estimates fewer parameters.
- **Performs well in practice:** Even though the independence assumption rarely holds true, Naive Bayes often works surprisingly well.

1.0.1 Disadvantages of Naive Bayes Classifier:

- **Strong independence assumption:** The assumption that features are conditionally independent given the class can lead to suboptimal results if this condition is not met in the data.
- **Zero probability problem:** If a feature value that wasn't seen in the training data appears in the test set, the model will assign a zero probability to that class. This can be mitigated using techniques like Laplace smoothing.
- **Not suitable for continuous data without modifications:** The basic form of Naive Bayes does not handle continuous data well unless it is adapted (e.g., Gaussian Naive Bayes).
- **Not ideal for highly correlated features:** Naive Bayes struggles when features are correlated since the independence assumption is violated.

Real-World Applications of Naive Bayes Classifier:

- **Email Spam Detection:** Classifies emails as spam or not based on the occurrence of certain words.
- **Text Classification:** Categorizes documents or news articles into different topics based on word occurrences (Multinomial Naive Bayes).
- **Sentiment Analysis:** Determines whether a review or a tweet is positive or negative by analyzing word frequencies (commonly used in NLP).
- **Medical Diagnosis:** Predicts diseases based on symptoms by calculating the probability of each condition.
- **Recommender Systems:** Helps in filtering and recommending content based on a user's previous interactions and preferences.

```
[55]: import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
```

```

from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import classification_report
from sklearn.model_selection import GridSearchCV

num_items = 41
data = {
    'City Population': np.random.randint(10000, 1000000, num_items),
    'Continent': np.random.choice(['Asia', 'Europe', 'North America', 'South_
    ↪America'], num_items),
    'Venue Capacity': np.random.randint(500, 20000, num_items),
    'Day Of Week': np.random.choice(['Monday', 'Tuesday', 'Wednesday',
    ↪Thursday', 'Friday', 'Saturday', 'Sunday'], num_items),
    'Multiple Concerts': np.random.randint(0, 2, num_items),
    'Sold Out': np.random.randint(0, 2, num_items)
}

df = pd.DataFrame(data)
#df = pd.read_csv('pear_jam_tour2.csv', encoding="unicode_escape")
df

```

```

[55]:
  City Population  Continent  Venue Capacity  Day Of Week  \
0          491817  North America          16665   Thursday
1          701437         Europe           9246    Monday
2          166930         Europe           8403   Thursday
3          588655  South America           4672   Saturday
4          388841  South America           1405    Friday
5          882813         Asia           18118    Sunday
6          158157         Asia            1666  Wednesday
7          117802         Asia          14382    Sunday
8           44163  North America           2502   Saturday
9          893911  North America          17897    Friday
10         981604         Europe          15859    Tuesday
11         780283  North America          19257  Wednesday
12         265129         Asia           4524   Thursday
13         131209         Asia          14857   Thursday
14         405167         Europe           9712    Monday
15          84701  North America           2482   Thursday
16         822257  North America           5842    Sunday
17         424221  South America          17013   Saturday
18         305672         Asia           2628    Sunday
19         720117  North America           3999   Thursday
20         285205  North America           6290   Saturday
21         121093  North America          10833   Thursday
22         240183         Asia           8535   Saturday
23         368908  North America          19451    Tuesday
24         989683  North America           8082   Saturday
25         743758  South America          10294    Monday

```

26	951668	Asia	15488	Friday
27	356715	Europe	1282	Wednesday
28	428998	South America	17548	Friday
29	490859	Europe	9096	Friday
30	815163	Asia	12759	Tuesday
31	568414	North America	3550	Friday
32	569051	South America	5245	Sunday
33	327147	Asia	4974	Wednesday
34	703689	North America	19689	Monday
35	368834	Europe	15419	Monday
36	259567	North America	16369	Tuesday
37	654837	North America	19461	Friday
38	560703	North America	19339	Thursday
39	466105	South America	12232	Tuesday
40	167668	South America	2890	Thursday

	Multiple Concerts	Sold Out
0	0	0
1	0	0
2	0	0
3	0	1
4	1	1
5	1	1
6	1	0
7	0	0
8	0	0
9	1	1
10	1	1
11	0	1
12	0	1
13	1	1
14	0	0
15	0	0
16	0	1
17	1	0
18	0	1
19	0	0
20	1	0
21	1	1
22	1	1
23	1	1
24	0	0
25	0	0
26	0	1
27	0	0
28	0	0
29	0	0

30	1	1
31	0	1
32	1	1
33	0	0
34	0	0
35	0	0
36	0	0
37	1	1
38	1	0
39	0	1
40	0	1

```
[56]: df2 = pd.get_dummies(df[['Continent', 'Day Of Week']])

df2
```

```
[56]:
```

	Continent_Asia	Continent_Europe	Continent_North America	\
0	False	False	True	
1	False	True	False	
2	False	True	False	
3	False	False	False	
4	False	False	False	
5	True	False	False	
6	True	False	False	
7	True	False	False	
8	False	False	True	
9	False	False	True	
10	False	True	False	
11	False	False	True	
12	True	False	False	
13	True	False	False	
14	False	True	False	
15	False	False	True	
16	False	False	True	
17	False	False	False	
18	True	False	False	
19	False	False	True	
20	False	False	True	
21	False	False	True	
22	True	False	False	
23	False	False	True	
24	False	False	True	
25	False	False	False	
26	True	False	False	
27	False	True	False	
28	False	False	False	
29	False	True	False	

30	True	False	False
31	False	False	True
32	False	False	False
33	True	False	False
34	False	False	True
35	False	True	False
36	False	False	True
37	False	False	True
38	False	False	True
39	False	False	False
40	False	False	False

	Continent_South America	Day Of Week_Friday	Day Of Week_Monday \
0	False	False	False
1	False	False	True
2	False	False	False
3	True	False	False
4	True	True	False
5	False	False	False
6	False	False	False
7	False	False	False
8	False	False	False
9	False	True	False
10	False	False	False
11	False	False	False
12	False	False	False
13	False	False	False
14	False	False	True
15	False	False	False
16	False	False	False
17	True	False	False
18	False	False	False
19	False	False	False
20	False	False	False
21	False	False	False
22	False	False	False
23	False	False	False
24	False	False	False
25	True	False	True
26	False	True	False
27	False	False	False
28	True	True	False
29	False	True	False
30	False	False	False
31	False	True	False
32	True	False	False
33	False	False	False

34	False	False	True
35	False	False	True
36	False	False	False
37	False	True	False
38	False	False	False
39	True	False	False
40	True	False	False

	Day Of Week_Saturday	Day Of Week_Sunday	Day Of Week_Thursday \
0	False	False	True
1	False	False	False
2	False	False	True
3	True	False	False
4	False	False	False
5	False	True	False
6	False	False	False
7	False	True	False
8	True	False	False
9	False	False	False
10	False	False	False
11	False	False	False
12	False	False	True
13	False	False	True
14	False	False	False
15	False	False	True
16	False	True	False
17	True	False	False
18	False	True	False
19	False	False	True
20	True	False	False
21	False	False	True
22	True	False	False
23	False	False	False
24	True	False	False
25	False	False	False
26	False	False	False
27	False	False	False
28	False	False	False
29	False	False	False
30	False	False	False
31	False	False	False
32	False	True	False
33	False	False	False
34	False	False	False
35	False	False	False
36	False	False	False
37	False	False	False

38	False	False	True
39	False	False	False
40	False	False	True

	Day Of Week_Tuesday	Day Of Week_Wednesday
0	False	False
1	False	False
2	False	False
3	False	False
4	False	False
5	False	False
6	False	True
7	False	False
8	False	False
9	False	False
10	True	False
11	False	True
12	False	False
13	False	False
14	False	False
15	False	False
16	False	False
17	False	False
18	False	False
19	False	False
20	False	False
21	False	False
22	False	False
23	True	False
24	False	False
25	False	False
26	False	False
27	False	True
28	False	False
29	False	False
30	True	False
31	False	False
32	False	False
33	False	True
34	False	False
35	False	False
36	True	False
37	False	False
38	False	False
39	True	False
40	False	False

```
[57]: df3 = pd.concat([df, df2], axis=1)
```

```
df3
```

```
[57]:
```

	City	Population	Continent	Venue	Capacity	Day Of Week	\
0		491817	North America		16665	Thursday	
1		701437	Europe		9246	Monday	
2		166930	Europe		8403	Thursday	
3		588655	South America		4672	Saturday	
4		388841	South America		1405	Friday	
5		882813	Asia		18118	Sunday	
6		158157	Asia		1666	Wednesday	
7		117802	Asia		14382	Sunday	
8		44163	North America		2502	Saturday	
9		893911	North America		17897	Friday	
10		981604	Europe		15859	Tuesday	
11		780283	North America		19257	Wednesday	
12		265129	Asia		4524	Thursday	
13		131209	Asia		14857	Thursday	
14		405167	Europe		9712	Monday	
15		84701	North America		2482	Thursday	
16		822257	North America		5842	Sunday	
17		424221	South America		17013	Saturday	
18		305672	Asia		2628	Sunday	
19		720117	North America		3999	Thursday	
20		285205	North America		6290	Saturday	
21		121093	North America		10833	Thursday	
22		240183	Asia		8535	Saturday	
23		368908	North America		19451	Tuesday	
24		989683	North America		8082	Saturday	
25		743758	South America		10294	Monday	
26		951668	Asia		15488	Friday	
27		356715	Europe		1282	Wednesday	
28		428998	South America		17548	Friday	
29		490859	Europe		9096	Friday	
30		815163	Asia		12759	Tuesday	
31		568414	North America		3550	Friday	
32		569051	South America		5245	Sunday	
33		327147	Asia		4974	Wednesday	
34		703689	North America		19689	Monday	
35		368834	Europe		15419	Monday	
36		259567	North America		16369	Tuesday	
37		654837	North America		19461	Friday	
38		560703	North America		19339	Thursday	
39		466105	South America		12232	Tuesday	
40		167668	South America		2890	Thursday	

	Multiple Concerts	Sold Out	Continent_Asia	Continent_Europe	\
0	0	0	False	False	
1	0	0	False	True	
2	0	0	False	True	
3	0	1	False	False	
4	1	1	False	False	
5	1	1	True	False	
6	1	0	True	False	
7	0	0	True	False	
8	0	0	False	False	
9	1	1	False	False	
10	1	1	False	True	
11	0	1	False	False	
12	0	1	True	False	
13	1	1	True	False	
14	0	0	False	True	
15	0	0	False	False	
16	0	1	False	False	
17	1	0	False	False	
18	0	1	True	False	
19	0	0	False	False	
20	1	0	False	False	
21	1	1	False	False	
22	1	1	True	False	
23	1	1	False	False	
24	0	0	False	False	
25	0	0	False	False	
26	0	1	True	False	
27	0	0	False	True	
28	0	0	False	False	
29	0	0	False	True	
30	1	1	True	False	
31	0	1	False	False	
32	1	1	False	False	
33	0	0	True	False	
34	0	0	False	False	
35	0	0	False	True	
36	0	0	False	False	
37	1	1	False	False	
38	1	0	False	False	
39	0	1	False	False	
40	0	1	False	False	

	Continent_North America	Continent_South America	Day Of Week_Friday	\
0	True	False	False	
1	False	False	False	
2	False	False	False	

3	False	True	False
4	False	True	True
5	False	False	False
6	False	False	False
7	False	False	False
8	True	False	False
9	True	False	True
10	False	False	False
11	True	False	False
12	False	False	False
13	False	False	False
14	False	False	False
15	True	False	False
16	True	False	False
17	False	True	False
18	False	False	False
19	True	False	False
20	True	False	False
21	True	False	False
22	False	False	False
23	True	False	False
24	True	False	False
25	False	True	False
26	False	False	True
27	False	False	False
28	False	True	True
29	False	False	True
30	False	False	False
31	True	False	True
32	False	True	False
33	False	False	False
34	True	False	False
35	False	False	False
36	True	False	False
37	True	False	True
38	True	False	False
39	False	True	False
40	False	True	False

	Day Of Week_Monday	Day Of Week_Saturday	Day Of Week_Sunday \
0	False	False	False
1	True	False	False
2	False	False	False
3	False	True	False
4	False	False	False
5	False	False	True
6	False	False	False

7	False	False	True
8	False	True	False
9	False	False	False
10	False	False	False
11	False	False	False
12	False	False	False
13	False	False	False
14	True	False	False
15	False	False	False
16	False	False	True
17	False	True	False
18	False	False	True
19	False	False	False
20	False	True	False
21	False	False	False
22	False	True	False
23	False	False	False
24	False	True	False
25	True	False	False
26	False	False	False
27	False	False	False
28	False	False	False
29	False	False	False
30	False	False	False
31	False	False	False
32	False	False	True
33	False	False	False
34	True	False	False
35	True	False	False
36	False	False	False
37	False	False	False
38	False	False	False
39	False	False	False
40	False	False	False

	Day Of Week_Thursday	Day Of Week_Tuesday	Day Of Week_Wednesday
0	True	False	False
1	False	False	False
2	True	False	False
3	False	False	False
4	False	False	False
5	False	False	False
6	False	False	True
7	False	False	False
8	False	False	False
9	False	False	False
10	False	True	False

11	False	False	True
12	True	False	False
13	True	False	False
14	False	False	False
15	True	False	False
16	False	False	False
17	False	False	False
18	False	False	False
19	True	False	False
20	False	False	False
21	True	False	False
22	False	False	False
23	False	True	False
24	False	False	False
25	False	False	False
26	False	False	False
27	False	False	True
28	False	False	False
29	False	False	False
30	False	True	False
31	False	False	False
32	False	False	False
33	False	False	True
34	False	False	False
35	False	False	False
36	False	True	False
37	False	False	False
38	True	False	False
39	False	True	False
40	True	False	False

```
[58]: df4 = df3.drop(columns=['Continent', 'Day Of Week'], axis=1)
df4
```

```
[58]:
```

	City	Population	Venue Capacity	Multiple Concerts	Sold Out	\
0		491817	16665	0	0	
1		701437	9246	0	0	
2		166930	8403	0	0	
3		588655	4672	0	1	
4		388841	1405	1	1	
5		882813	18118	1	1	
6		158157	1666	1	0	
7		117802	14382	0	0	
8		44163	2502	0	0	
9		893911	17897	1	1	
10		981604	15859	1	1	
11		780283	19257	0	1	

12	265129	4524	0	1
13	131209	14857	1	1
14	405167	9712	0	0
15	84701	2482	0	0
16	822257	5842	0	1
17	424221	17013	1	0
18	305672	2628	0	1
19	720117	3999	0	0
20	285205	6290	1	0
21	121093	10833	1	1
22	240183	8535	1	1
23	368908	19451	1	1
24	989683	8082	0	0
25	743758	10294	0	0
26	951668	15488	0	1
27	356715	1282	0	0
28	428998	17548	0	0
29	490859	9096	0	0
30	815163	12759	1	1
31	568414	3550	0	1
32	569051	5245	1	1
33	327147	4974	0	0
34	703689	19689	0	0
35	368834	15419	0	0
36	259567	16369	0	0
37	654837	19461	1	1
38	560703	19339	1	0
39	466105	12232	0	1
40	167668	2890	0	1

	Continent_Asia	Continent_Europe	Continent_North America	\
0	False	False	True	
1	False	True	False	
2	False	True	False	
3	False	False	False	
4	False	False	False	
5	True	False	False	
6	True	False	False	
7	True	False	False	
8	False	False	True	
9	False	False	True	
10	False	True	False	
11	False	False	True	
12	True	False	False	
13	True	False	False	
14	False	True	False	
15	False	False	True	

16	False	False	True
17	False	False	False
18	True	False	False
19	False	False	True
20	False	False	True
21	False	False	True
22	True	False	False
23	False	False	True
24	False	False	True
25	False	False	False
26	True	False	False
27	False	True	False
28	False	False	False
29	False	True	False
30	True	False	False
31	False	False	True
32	False	False	False
33	True	False	False
34	False	False	True
35	False	True	False
36	False	False	True
37	False	False	True
38	False	False	True
39	False	False	False
40	False	False	False

	Continent_South America	Day Of Week_Friday	Day Of Week_Monday \
0	False	False	False
1	False	False	True
2	False	False	False
3	True	False	False
4	True	True	False
5	False	False	False
6	False	False	False
7	False	False	False
8	False	False	False
9	False	True	False
10	False	False	False
11	False	False	False
12	False	False	False
13	False	False	False
14	False	False	True
15	False	False	False
16	False	False	False
17	True	False	False
18	False	False	False
19	False	False	False

20	False	False	False
21	False	False	False
22	False	False	False
23	False	False	False
24	False	False	False
25	True	False	True
26	False	True	False
27	False	False	False
28	True	True	False
29	False	True	False
30	False	False	False
31	False	True	False
32	True	False	False
33	False	False	False
34	False	False	True
35	False	False	True
36	False	False	False
37	False	True	False
38	False	False	False
39	True	False	False
40	True	False	False

	Day Of Week_Saturday	Day Of Week_Sunday	Day Of Week_Thursday	\
0	False	False	True	
1	False	False	False	
2	False	False	True	
3	True	False	False	
4	False	False	False	
5	False	True	False	
6	False	False	False	
7	False	True	False	
8	True	False	False	
9	False	False	False	
10	False	False	False	
11	False	False	False	
12	False	False	True	
13	False	False	True	
14	False	False	False	
15	False	False	True	
16	False	True	False	
17	True	False	False	
18	False	True	False	
19	False	False	True	
20	True	False	False	
21	False	False	True	
22	True	False	False	
23	False	False	False	

24	True	False	False
25	False	False	False
26	False	False	False
27	False	False	False
28	False	False	False
29	False	False	False
30	False	False	False
31	False	False	False
32	False	True	False
33	False	False	False
34	False	False	False
35	False	False	False
36	False	False	False
37	False	False	False
38	False	False	True
39	False	False	False
40	False	False	True

	Day Of Week_Tuesday	Day Of Week_Wednesday
0	False	False
1	False	False
2	False	False
3	False	False
4	False	False
5	False	False
6	False	True
7	False	False
8	False	False
9	False	False
10	True	False
11	False	True
12	False	False
13	False	False
14	False	False
15	False	False
16	False	False
17	False	False
18	False	False
19	False	False
20	False	False
21	False	False
22	False	False
23	True	False
24	False	False
25	False	False
26	False	False
27	False	True

28	False	False
29	False	False
30	True	False
31	False	False
32	False	False
33	False	True
34	False	False
35	False	False
36	True	False
37	False	False
38	False	False
39	True	False
40	False	False

```
[59]: X = df4.drop(columns=['Sold Out'], axis=1)
      y = df4['Sold Out']

      X_train, X_test, y_train, y_test = train_test_split(X,y, random_state=33,
      ↪test_size=0.2)
      gnb = GaussianNB()

      gnb.fit(X_train, y_train)
      y_pred = gnb.predict(X_test)
      print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.80	0.80	0.80	5
1	0.75	0.75	0.75	4
accuracy			0.78	9
macro avg	0.78	0.78	0.78	9
weighted avg	0.78	0.78	0.78	9

```
[60]: gnb.score(X_train, y_train)
```

```
[60]: 0.5625
```

```
[61]: gnb.score(X_test, y_test)
```

```
[61]: 0.7777777777777778
```

1.0.2 Add in parameter

```
[66]: param_grid = {  
      'var_smoothing': [0.00000001, 0.000000001, 0.00000001]  
      }  
  
      grid_search = GridSearchCV(gnb,param_grid, cv=5, scoring='accuracy', n_jobs=-1)  
      grid_search.fit(X_train, y_train)
```

```
[66]: GridSearchCV(cv=5, estimator=GaussianNB(), n_jobs=-1,  
                  param_grid={'var_smoothing': [1e-08, 1e-09, 1e-08]},  
                  scoring='accuracy')
```

```
[67]: grid_search.best_params_
```

```
[67]: {'var_smoothing': 1e-08}
```

```
[68]: grid_search.best_score_
```

```
[68]: 0.4095238095238095
```